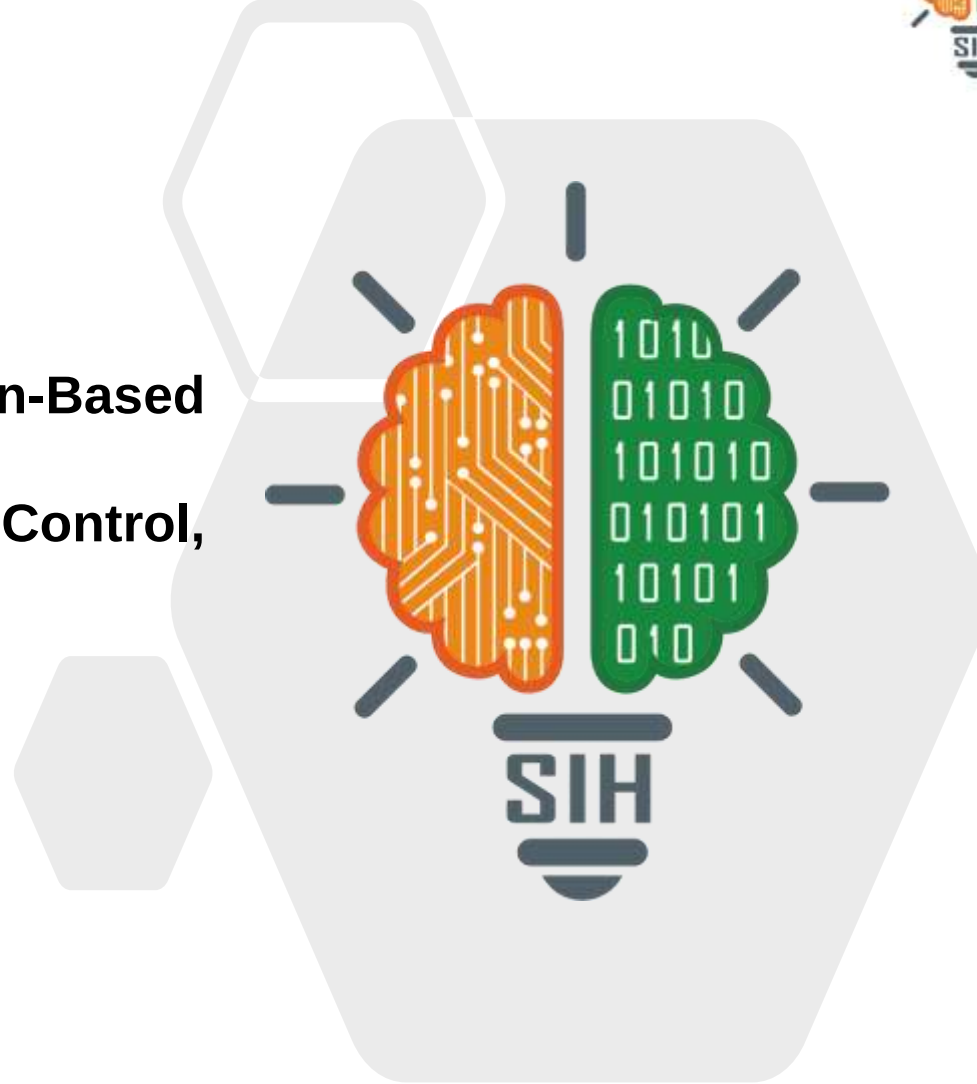


SMART INDIA HACKATHON 2026



- Problem Statement ID –26125
- Problem Statement Title - Blockchain-Based Secure Platform for Identity, Access Control, and Digital Asset Management
- Theme - Blockchain & Cybersecurity
- PS Category - Software
- Team ID – GLAS51
- Team Name - TrustMesh



TrustMesh - A Blockchain-Based Trust Layer for Identity, Access Control & Digital Assets

❖ The Problem

- Centralized identity systems create security risks and single points of failure.
- Identity theft & unauthorized access are difficult to prevent and verify.
- Digital assets and access rights are fragmented across disconnected systems.
- Ownership, access rights & authenticity are difficult to track and vulnerable to tampering.

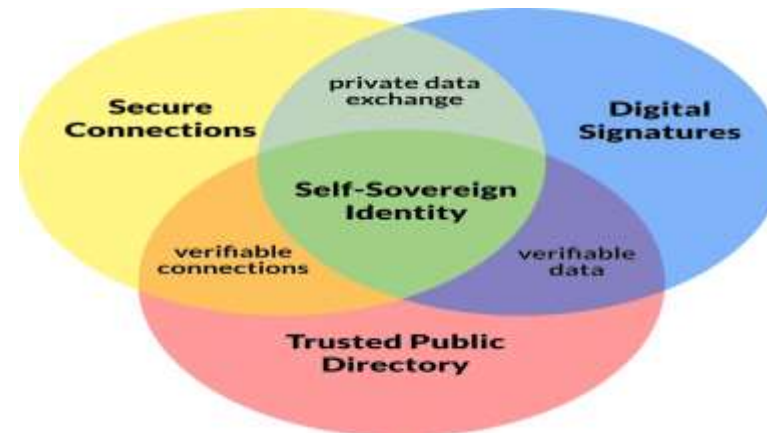
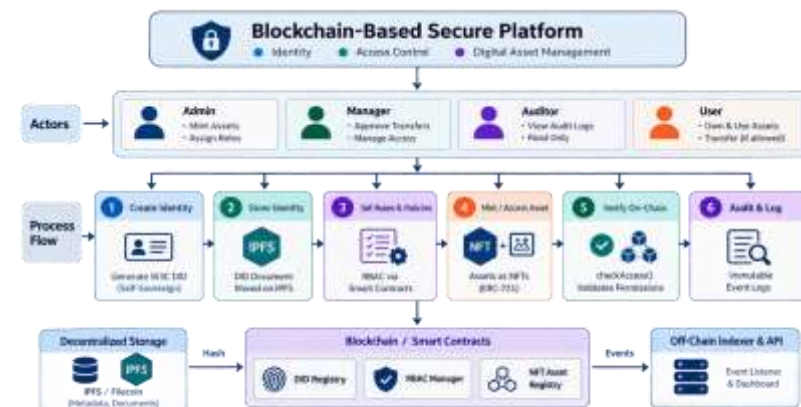
❖ The Proposed Solution

- A reusable Trust Layer connects applications to blockchain-backed identity, policy enforcement, asset ownership and audit evidence..
- DID-based identity provides a cryptographically verifiable identity layer for participating applications.
- NFT-based digital assets provide verifiable ownership and controlled transfers through smart contracts.
- PolicyEngine enforces access decisions on-chain, while AuditLogger records trust-relevant actions as verifiable evidence..

❖ Impact and Innovation

Reduces dependence on application-specific authorization by introducing a common trust layer.

- Blockchain-backed asset records make ownership and authorized transfers verifiable.
- Smart-contract enforcement prevents unauthorized access and permission bypass.
- Generalized Policy Engine supports NFTs, documents, APIs and physical access without redeployment.
- Multi-tenant architecture allows multiple organizations to securely share the same platform.
- Guardian-based social recovery enables secure credential recovery with time-lock protection.





❖ Technologies Used

SMART CONTRACTS

- Solidity+OpenZeppelin AccessControl,ERC-721

BLOCK CHAIN

- Polygon for MVP/demo
Hyperledger Fabric for production

IDENTITY

- W3C DID(did:ethr/did:key,did-jwt)

DECENTRALIZED STORAGE

- IPFS via Pinata/Web3.Storage Metadata & documents

BACKEND

- [Node.js](#) + Express/NestJS,Off-chain event indexer

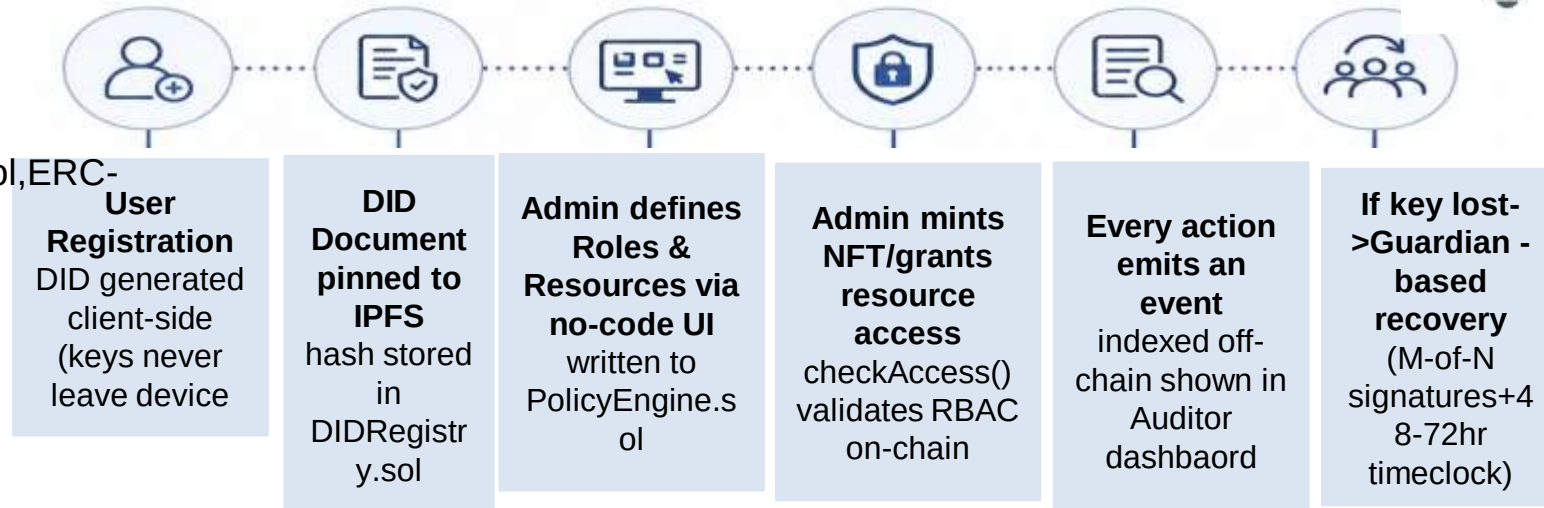
FRONTEND

- React+[ethers.js/NestJS](#) Off-chain event indexer

DEV/TEST

- Hardhat Local Chain,testing & deployment

❖ Methodology & Process



❖ System-Architecture



❖ Prototype Plan



Key Tools & Logos



Potential Challenges & Risks

-  Lost private keys causing identity loss, with no central reset option available.
-  High gas costs on public chains, making frequent operations expensive and limiting scalability.
-  Immutable data conflicting with DPDP Act requirements, creating compliance issues with erasure rights.
-  Admin key compromise enabling fake asset minting or unauthorized role assignment in the system.
-  Low crypto-literacy creating friction in onboarding, as seed phrases and fees confuse mainstream users.

Strategies to Overcome Challenges

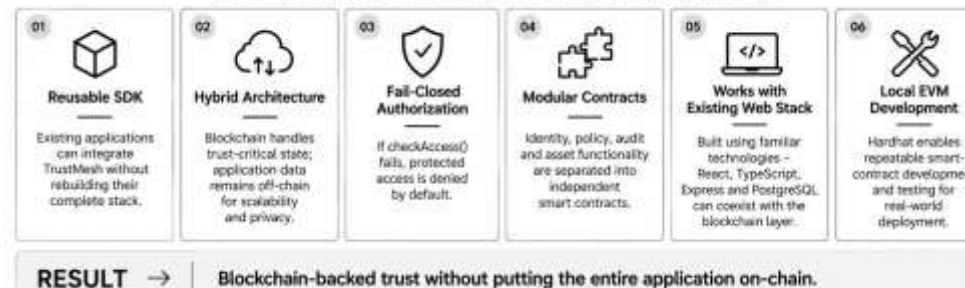
-  Guardian-based recovery with institutional fallback ensures identity restoration even if keys are lost.
-  L2 or permissioned deployment reduces transaction costs and scales more efficiently for enterprises.
-  Off-chain encrypted PII with only hashes on-chain enables lawful erasure while preserving integrity.
-  Multisig and hardware wallets strengthen admin security, preventing single-key compromise risks.
-  Account abstraction (ERC-4337) simplifies onboarding by hiding blockchain complexity and sponsoring gas.

Feasibility Analysis



WHY TRUSTMESH STANDS OUT

A practical, secure and scalable trust infrastructure for real-world applications.

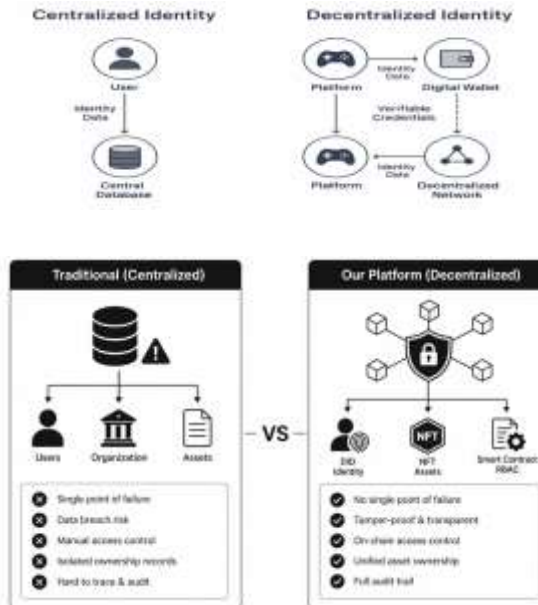


WHY THIS IS NEEDED

2025 evidence behind the need for stronger identity & access infrastructure



PROCAP - IBM India's Data Breach Report 2026 - IBM India Newsroom, 09 Aug 2025



Government

Tamper-proof identity & record systems (land, certificates, licenses) → prevents forgery and unauthorized changes by keeping records immutable and verifiable.

Enterprises

Unified IAM + asset custody + compliance logging. → Eliminates siloed systems, reducing fraud and simplifying audits with one secure platform.

Citizens

Self-sovereign identity, no lockouts or breaches → ensures individuals control their digital identity, protecting against misuse and central lockouts.

Auditors

Real-time, verifiable audit trails for accountability → provides indisputable evidence instantly, cutting disputes and speeding compliance checks.

Economic

Cuts fraud losses & manual verification costs → reduces operational overhead while improving trust in digital transactions.

Scalability

One policy engine serving multiple domains (education, healthcare, supply chain, land records) → maximizes impact with minimal redeployment effort.

Interoperability

Portable W3C DID identities across platforms → allows seamless use of one identity across multiple apps and ecosystems.

Security

Eliminates the single point of failure in centralized IAM → ensures identities and assets remain safe even if one node is compromised.

**MARKET POTENTIAL**

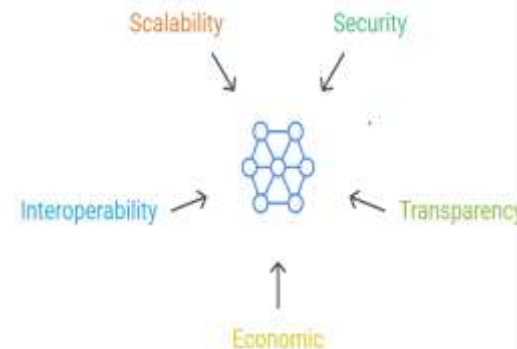
Global IAM Market
\$34.3B
(2024)
CAGR 12.6%
(2024–2030)



India Cybersecurity Market
\$4.3B (2024)
CAGR 13.6%
(2024–2030)



Huge opportunity for decentralized identity and access management in India.

**FUTURE SCOPE & ESTIMATES**

1. GROWTH ESTIMATES (USERS)		
Year	Estimated Users	Growth
2025	50K+	Early Adopters
2026	2L+	Expanding Across Sectors
2027	10L+	Wider Adoption
2028	25L+	Strong Network Effect
2029	50L+	Bharat-wide Trust Layer

2. FUTURE SCOPE BY DOMAIN		
Domain	Future Scope	Benefit
Education	Degree & certificate verification	Eliminate fake certificates, easy verification
Healthcare	Secure access to medical records	Patient-controlled data sharing
Government	Licenses, certificates, citizen services	Faster, transparent & tamper-proof
Land / Assets	Property & asset ownership	Prevent fraud, verify real ownership
Supply Chain	Product tracking & provenance	End-to-end transparency
Enterprise / IAM	Access management & compliance	Secure roles, permissions & audit

3. 3-YEAR ROADMAP			
Phase	Timeline	Focus Areas	Outcomes
Phase 1	2025	Build Core Platform, Onboard Early Partners	Strong Foundation
Phase 2	2026	Integrate More Sectors, APIs & Systems	Expansion & Integration
Phase 3	2027	Mass Adoption Across Bharat	Become Default Trust Layer

4. VISION 2030	
To become the foundational trust infrastructure for every identity, asset & transaction in Bharat.	

1. W3C Decentralized Identifiers v1.1 – Candidate Rec. 2026 –Defines global, self-sovereign digital identities without central control.

2. Hyperledger Fabric Docs – Enterprise-grade permissioned blockchain for secure, modular networks.

1.

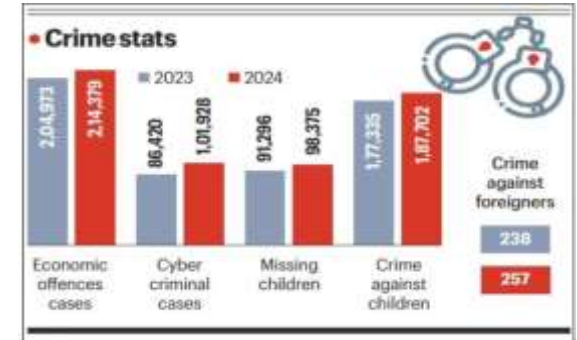
2.

3.

4.

OpenZeppelin Contracts (AccessControl, ERC-721) –Audited smart-contract library for RBAC and NFT standards.

India's DPDP Act, 2023 –Privacy law enforcing consent, user rights, and strict data compliance.



ASSISTANCE TO STATES TO TACKLE CYBER INCIDENTS

Project 101, 1st term 2024-25 (2024-25)

The India Cyber Security Response Team (ICSRT) is dedicated to the prompt response to cyber security incidents, under the purview of section 100 of the Information Technology Act, 2008. As per the information reported to ICSRT, the total number of cyber security incidents reported during the 1st term are given below:

Year	Number of cyber security incidents
2021	1,100,000
2022	1,100,000
2023	1,100,000
2024	1,100,000
2025	1,100,000

Ministry of Electronics & IT



Digital Public Infrastructure Strengthens Inclusive Governance by Expanding Access to Digital Services, Financial Inclusion and Last-Mile Service Delivery

Over 144 Crore Aadhaar IDs, 55.49 Crore UPI Users, 71.66 Crore DigiLocker Users and Over 5 Lakh Common Service Centres Power India's Digital Public Infrastructure

Digital India Bridges the Digital Divide with Over 2.21 Lakh BharatNet-Enabled Gram Panchayats, Over 800 Services through Common Service Centres and 6.39 Crore People Trained in Digital Literacy

10:00 PM, 29 JUL 2025 4:50PM by PIB Delhi

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Business News | News | India | How a Btech dropout floated fake passport and UIDAI portals to tamper with data of over 1,500 Aadhaar cardholders across 12 states

How a BTech dropout floated fake passport and UIDAI portals to tamper with data of over 1,500 Aadhaar cardholders across 12 states

This story is from April 07, 2025

How UP-based gang tampered Aadhaar biometrics, altered thousands of digital identities across 12 states

Sandeep Rai / TNN / Updated: Apr 08, 2025, 15:26 IST

Share Print

A hi-tech identity fraud racket, operating from Uttar Pradesh, has been busted for allegedly tampering with biometric data of over 1,500 Aadhaar cardholders across 12 states. Police arrested four individuals who exploited loopholes in the UIDAI system, running a network of agents who collected documents for illegal updates.



MEERUT: UP police have uncovered a hi-tech identity fraud racket operating from Badaun and Amroha allegedly responsible for "tampering with biometric data of over 1,500 Aadhaar cardholders across 12 states". A joint cyber team, supervised by SP Sambhal Krishna Kumar Bishnoi and ASP Anukriti Sharma, busted the operation and arrested four "key players".